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United States Patent	12401116
Kind Code	B2
Date of Patent	August 26, 2025
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Electronic device with multi-feed antenna structures

Abstract

An electronic device may be provided with peripheral conductive housing structures having a first segment and a second segment. First and second antenna feeds may be coupled between the first segment and the ground structures. The first feed may convey signals in a first band and the second feed may convey signals in a second band. The first segment may be near-field coupled to a slot between the second segment and the ground structures. A first tuner may be coupled between the second segment and the ground structures and may adjust a resonance of the first segment in the first and second bands. A second tuner coupled to the first feed may perform impedance matching in the first band and aperture tuning in the second band. A third tuner coupled to the second feed may perform impedance matching in the second band and aperture tuning in the first band.

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Appl. No.: 18/458949

Filed: August 30, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240079766 A1	Mar. 07, 2024

Related U.S. Application Data

Publication Classification**Int. Cl.:** **H01Q1/38** (20060101); **H01Q1/50** (20060101); **H01Q13/10** (20060101)**U.S. Cl.:****CPC** **H01Q1/38** (20130101); **H01Q1/50** (20130101); **H01Q13/10** (20130101);**Field of Classification Search****CPC:** H01Q (1/38); H01Q (1/50); H01Q (13/10); H01Q (5/328); H01Q (5/335); H01Q (9/42); H01Q (1/243)

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Background/Summary

(1) This application claims the benefit of U.S. Provisional Patent Application No. 63/403,977, filed Sep. 6, 2022, which is hereby incorporated by reference herein in its entirety.

BACKGROUND

(1) This relates generally to electronic devices and, more particularly, to electronic devices with wireless communications capabilities.

(2) Electronic devices such as portable computers and cellular telephones are often provided with wireless communications capabilities. To satisfy consumer demand for small form factor wireless devices, manufacturers are continually striving to implement wireless communications circuitry such as antenna components using compact structures. At the same time, there is a desire for wireless devices to cover a growing number of communications bands.

(3) Because antennas have the potential to interfere with each other and with components in a

wireless device, care must be taken when incorporating antennas into an electronic device.

Moreover, care must be taken to ensure that the antennas and wireless circuitry in a device are able to exhibit satisfactory performance over a range of operating frequencies and with satisfactory efficiency bandwidth.

SUMMARY

(4) An electronic device may be provided with wireless circuitry and a housing having peripheral conductive housing structures. A dielectric-filled gap may divide the peripheral conductive housing structures into a first segment and a second segment. A first antenna feed may be coupled between the first segment and ground structures. A second antenna feed may be coupled between the first segment and the ground structures. The first antenna feed may convey radio-frequency signals in a first frequency band. The second antenna feed may convey radio-frequency signals in a second frequency band.

(5) The first segment may be near-field coupled to a slot between the second segment and the ground structures in the first and second frequency bands. A first tuner may be coupled between the second segment and the ground structures across the slot. The first tuner may adjust a resonance of the first segment in the first and second frequency bands. If desired, the first tuner may also perform impedance matching for a third antenna feed coupled between the second segment and the ground structures across the slot.

(6) A second tuner may be coupled to the first antenna feed. A third tuner may be coupled to the second antenna feed. The second tuner may perform impedance matching in the first frequency band for the first antenna. The second tuner may also perform aperture tuning in the second frequency band for the second antenna. The third tuner may perform impedance matching in the second frequency band for the second antenna. The third tuner may also perform aperture tuning in the first frequency band for the first antenna.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an illustrative electronic device in accordance with some embodiments.

(2) FIG. 2 is a schematic diagram of illustrative circuitry in an electronic device in accordance with some embodiments.

(3) FIG. 3 is a schematic diagram of illustrative wireless circuitry in accordance with some embodiments.

(4) FIG. 4 is a cross-sectional side view of an illustrative electronic device having housing structures that may be used in forming antenna structures in accordance with some embodiments.

(5) FIG. 5 is an interior front view of the lower end of an illustrative electronic device having peripheral conductive housing structures with dielectric gaps for forming at least first and second antennas in accordance with some embodiments.

(6) FIG. 6 is a schematic diagram showing how conductive structures in an antenna may be near-field coupled to conductive structures in one or more other antennas for optimizing performance of the antennas in accordance with some embodiments.

(7) FIG. 7 is an interior rear view of first and second antennas that share near-field coupled conductive structures in accordance with some embodiments.

(8) FIG. 8 is an interior rear view showing how a third antenna may be integrated into the conductive structures of first and second antennas of the type shown in FIG. 7 in accordance with some embodiments.

DETAILED DESCRIPTION

(9) An electronic device such as electronic device **10** of FIG. 1 may be provided with wireless

circuitry that includes antennas. The antennas may be used to transmit and/or receive wireless radio-frequency signals.

(10) Device **10** may be a portable electronic device or other suitable electronic device. For example, device **10** may be a laptop computer, a tablet computer, a somewhat smaller device such as a wrist-watch device, pendant device, headphone device, earpiece device, headset device, or other wearable or miniature device, a handheld device such as a cellular telephone, a media player, or other small portable device. Device **10** may also be a set-top box, a desktop computer, a display into which a computer or other processing circuitry has been integrated, a display without an integrated computer, a wireless access point, a wireless base station, an electronic device incorporated into a kiosk, building, or vehicle, or other suitable electronic equipment.

(11) Device **10** may include a housing such as housing **12**. Housing **12**, which may sometimes be referred to as a case, may be formed of plastic, glass, ceramics, fiber composites, metal (e.g., stainless steel, aluminum, etc.), other suitable materials, or a combination of these materials. In some situations, parts of housing **12** may be formed from dielectric or other low-conductivity material (e.g., glass, ceramic, plastic, sapphire, etc.). In other situations, housing **12** or at least some of the structures that make up housing **12** may be formed from metal elements.

(12) Device **10** may, if desired, have a display such as display **14**. Display **14** may be mounted on the front face of device **10**. Display **14** may be a touch screen that incorporates capacitive touch electrodes or may be insensitive to touch. The rear face of housing **12** (i.e., the face of device **10** opposing the front face of device **10**) may have a substantially planar housing wall such as rear housing wall **12R** (e.g., a planar housing wall). Rear housing wall **12R** may have slots that pass entirely through the rear housing wall and that therefore separate portions of housing **12** from each other. Rear housing wall **12R** may include conductive portions and/or dielectric portions. If desired, rear housing wall **12R** may include a planar metal layer covered by a thin layer or coating of dielectric such as glass, plastic, sapphire, or ceramic (e.g., a dielectric cover layer). Housing **12** may also have shallow grooves that do not pass entirely through housing **12**. The slots and grooves may be filled with plastic or other dielectric materials. If desired, portions of housing **12** that have been separated from each other (e.g., by a through slot) may be joined by internal conductive structures (e.g., sheet metal or other metal members that bridge the slot).

(13) Housing **12** may include peripheral housing structures such as peripheral structures **12W**. Conductive portions of peripheral structures **12W** and conductive portions of rear housing wall **12R** may sometimes be referred to herein collectively as conductive structures of housing **12**. Peripheral structures **12W** may run around the periphery of device **10** and display **14**. In configurations in which device **10** and display **14** have a rectangular shape with four edges, peripheral structures **12W** may be implemented using peripheral housing structures that have a rectangular ring shape with four corresponding edges and that extend from rear housing wall **12R** to the front face of device **10** (as an example). In other words, device **10** may have a length (e.g., measured parallel to the Y-axis), a width that is less than the length (e.g., measured parallel to the X-axis), and a height (e.g., measured parallel to the Z-axis) that is less than the width. Peripheral structures **12W** or part of peripheral structures **12W** may serve as a bezel for display **14** (e.g., a cosmetic trim that surrounds all four sides of display **14** and/or that helps hold display **14** to device **10**) if desired. Peripheral structures **12W** may, if desired, form sidewall structures for device **10** (e.g., by forming a metal band with vertical sidewalls, curved sidewalls, etc.).

(14) Peripheral structures **12W** may be formed from a conductive material such as metal and may therefore sometimes be referred to as peripheral conductive housing structures, conductive housing structures, peripheral metal structures, peripheral conductive sidewalls, peripheral conductive sidewall structures, conductive housing sidewalls, peripheral conductive housing sidewalls, sidewalls, sidewall structures, or a peripheral conductive housing member (as examples). Peripheral conductive housing structures **12W** may be formed from a metal such as stainless steel, aluminum, alloys, or other suitable materials. One, two, or more than two separate structures may

be used in forming peripheral conductive housing structures **12W**.

(15) It is not necessary for peripheral conductive housing structures **12W** to have a uniform cross-section. For example, the top portion of peripheral conductive housing structures **12W** may, if desired, have an inwardly protruding ledge that helps hold display **14** in place. The bottom portion of peripheral conductive housing structures **12W** may also have an enlarged lip (e.g., in the plane of the rear surface of device **10**). Peripheral conductive housing structures **12W** may have substantially straight vertical sidewalls, may have sidewalls that are curved, or may have other suitable shapes. In some configurations (e.g., when peripheral conductive housing structures **12W** serve as a bezel for display **14**), peripheral conductive housing structures **12W** may run around the lip of housing **12** (i.e., peripheral conductive housing structures **12W** may cover only the edge of housing **12** that surrounds display **14** and not the rest of the sidewalls of housing **12**).

(16) Rear housing wall **12R** may lie in a plane that is parallel to display **14**. In configurations for device **10** in which some or all of rear housing wall **12R** is formed from metal, it may be desirable to form parts of peripheral conductive housing structures **12W** as integral portions of the housing structures forming rear housing wall **12R**. For example, rear housing wall **12R** of device **10** may include a planar metal structure and portions of peripheral conductive housing structures **12W** on the sides of housing **12** may be formed as flat or curved vertically extending integral metal portions of the planar metal structure (e.g., housing structures **12R** and **12W** may be formed from a continuous piece of metal in a unibody configuration). Housing structures such as these may, if desired, be machined from a block of metal and/or may include multiple metal pieces that are assembled together to form housing **12**. Rear housing wall **12R** may have one or more, two or more, or three or more portions. Peripheral conductive housing structures **12W** and/or conductive portions of rear housing wall **12R** may form one or more exterior surfaces of device **10** (e.g., surfaces that are visible to a user of device **10**) and/or may be implemented using internal structures that do not form exterior surfaces of device **10** (e.g., conductive housing structures that are not visible to a user of device **10** such as conductive structures that are covered with layers such as thin cosmetic layers, protective coatings, and/or other coating/cover layers that may include dielectric materials such as glass, ceramic, plastic, or other structures that form the exterior surfaces of device **10** and/or serve to hide peripheral conductive housing structures **12W** and/or conductive portions of rear housing wall **12R** from view of the user).

(17) Display **14** may have an array of pixels that form an active area **AA** that displays images for a user of device **10**. For example, active area **AA** may include an array of display pixels. The array of pixels may be formed from liquid crystal display (LCD) components, an array of electrophoretic pixels, an array of plasma display pixels, an array of organic light-emitting diode display pixels or other light-emitting diode pixels, an array of electrowetting display pixels, or display pixels based on other display technologies. If desired, active area **AA** may include touch sensors such as touch sensor capacitive electrodes, force sensors, or other sensors for gathering a user input.

(18) Display **14** may have an inactive border region that runs along one or more of the edges of active area **AA**. Inactive area **IA** of display **14** may be free of pixels for displaying images and may overlap circuitry and other internal device structures in housing **12**. To block these structures from view by a user of device **10**, the underside of the display cover layer or other layers in display **14** that overlap inactive area **IA** may be coated with an opaque masking layer in inactive area **IA**. The opaque masking layer may have any suitable color. Inactive area **IA** may include a recessed region such as notch **24** that extends into active area **AA**. Active area **AA** may, for example, be defined by the lateral area of a display module for display **14** (e.g., a display module that includes pixel circuitry, touch sensor circuitry, etc.). The display module may have a recess or notch in upper region **20** of device **10** that is free from active display circuitry (i.e., that forms notch **24** of inactive area **IA**). Notch **24** may be a substantially rectangular region that is surrounded (defined) on three sides by active area **AA** and on a fourth side by peripheral conductive housing structures **12W**. One or more sensors may be aligned with notch **24** and may transmit and/or receive light through

display **14** within notch **24**.

(19) Display **14** may be protected using a display cover layer such as a layer of transparent glass, clear plastic, transparent ceramic, sapphire, or other transparent crystalline material, or other transparent layer(s). The display cover layer may have a planar shape, a convex curved profile, a shape with planar and curved portions, a layout that includes a planar main area surrounded on one or more edges with a portion that is bent out of the plane of the planar main area, or other suitable shapes. The display cover layer may cover the entire front face of device **10**. In another suitable arrangement, the display cover layer may cover substantially all of the front face of device **10** or only a portion of the front face of device **10**. Openings may be formed in the display cover layer. For example, an opening may be formed in the display cover layer to accommodate a button. An opening may also be formed in the display cover layer to accommodate ports such as speaker port **16** in notch **24** or a microphone port. Openings may be formed in housing **12** to form communications ports (e.g., an audio jack port, a digital data port, etc.) and/or audio ports for audio components such as a speaker and/or a microphone if desired.

(20) Display **14** may include conductive structures such as an array of capacitive electrodes for a touch sensor, conductive lines for addressing pixels, driver circuits, etc. Housing **12** may include internal conductive structures such as metal frame members and a planar conductive housing member (sometimes referred to as a conductive support plate or backplate) that spans the walls of housing **12** (e.g., a substantially rectangular sheet formed from one or more metal parts that is welded or otherwise connected between opposing sides of peripheral conductive housing structures **12W**). The conductive support plate may form an exterior rear surface of device **10** or may be covered by a dielectric cover layer such as a thin cosmetic layer, protective coating, and/or other coatings that may include dielectric materials such as glass, ceramic, plastic, or other structures that form the exterior surfaces of device **10** and/or serve to hide the conductive support plate from view of the user (e.g., the conductive support plate may form part of rear housing wall **12R**). Device **10** may also include conductive structures such as printed circuit boards, components mounted on printed circuit boards, and other internal conductive structures. These conductive structures, which may be used in forming a ground plane in device **10**, may extend under active area AA of display **14**, for example.

(21) In regions **22** and **20**, openings may be formed within the conductive structures of device **10** (e.g., between peripheral conductive housing structures **12W** and opposing conductive ground structures such as conductive portions of rear housing wall **12R**, conductive traces on a printed circuit board, conductive electrical components in display **14**, etc.). These openings, which may sometimes be referred to as gaps, may be filled with air, plastic, and/or other dielectrics and may be used in forming slot antenna resonating elements for one or more antennas in device **10**, if desired.

(22) Conductive housing structures and other conductive structures in device **10** may serve as a ground plane for the antennas in device **10**. The openings in regions **22** and **20** may serve as slots in open or closed slot antennas, may serve as a central dielectric region that is surrounded by a conductive path of materials in a loop antenna, may serve as a space that separates an antenna resonating element such as a strip antenna resonating element or an inverted-F antenna resonating element from the ground plane, may contribute to the performance of a parasitic antenna resonating element, or may otherwise serve as part of antenna structures formed in regions **22** and **20**. If desired, the ground plane that is under active area AA of display **14** and/or other metal structures in device **10** may have portions that extend into parts of the ends of device **10** (e.g., the ground may extend towards the dielectric-filled openings in regions **22** and **20**), thereby narrowing the slots in regions **22** and **20**. Region **22** may sometimes be referred to herein as lower region **22** or lower end **22** of device **10**. Region **20** may sometimes be referred to herein as upper region **20** or upper end **20** of device **10**.

(23) In general, device **10** may include any suitable number of antennas (e.g., one or more, two or more, three or more, four or more, etc.). The antennas in device **10** may be located at opposing first

and second ends of an elongated device housing (e.g., at lower region **22** and/or upper region **20** of device **10** of FIG. **1**), along one or more edges of a device housing, in the center of a device housing, in other suitable locations, or in one or more of these locations. The arrangement of FIG. **1** is merely illustrative.

(24) Portions of peripheral conductive housing structures **12W** may be provided with peripheral gap structures. For example, peripheral conductive housing structures **12W** may be provided with one or more dielectric-filled gaps such as gaps **18**, as shown in FIG. **1**. The gaps in peripheral conductive housing structures **12W** may be filled with dielectric such as polymer, ceramic, glass, air, other dielectric materials, or combinations of these materials. Gaps **18** may divide peripheral conductive housing structures **12W** into one or more peripheral conductive segments. The conductive segments that are formed in this way may form parts of antennas in device **10** if desired. Other dielectric openings may be formed in peripheral conductive housing structures **12W** (e.g., dielectric openings other than gaps **18**) and may serve as dielectric antenna windows for antennas mounted within the interior of device **10**. Antennas within device **10** may be aligned with the dielectric antenna windows for conveying radio-frequency signals through peripheral conductive housing structures **12W**. Antennas within device **10** may also be aligned with inactive area **IA** of display **14** for conveying radio-frequency signals through display **14**.

(25) To provide an end user of device **10** with as large of a display as possible (e.g., to maximize an area of the device used for displaying media, running applications, etc.), it may be desirable to increase the amount of area at the front face of device **10** that is covered by active area **AA** of display **14**. Increasing the size of active area **AA** may reduce the size of inactive area **IA** within device **10**. This may reduce the area behind display **14** that is available for antennas within device **10**. For example, active area **AA** of display **14** may include conductive structures that serve to block radio-frequency signals handled by antennas mounted behind active area **AA** from radiating through the front face of device **10**. It would therefore be desirable to be able to provide antennas that occupy a small amount of space within device **10** (e.g., to allow for as large of a display active area **AA** as possible) while still allowing the antennas to communicate with wireless equipment external to device **10** with satisfactory efficiency bandwidth.

(26) In a typical scenario, device **10** may have one or more upper antennas and one or more lower antennas. An upper antenna may, for example, be formed in upper region **20** of device **10**. A lower antenna may, for example, be formed in lower region **22** of device **10**. Additional antennas may be formed along the edges of housing **12** extending between regions **20** and **22** if desired. The antennas may be used separately to cover identical communications bands, overlapping communications bands, or separate communications bands. The antennas may be used to implement an antenna diversity scheme or a multiple-input-multiple-output (MIMO) antenna scheme. Other antennas for covering any other desired frequencies may also be mounted at any desired locations within the interior of device **10**. The example of FIG. **1** is merely illustrative. If desired, housing **12** may have other shapes (e.g., a square shape, cylindrical shape, spherical shape, combinations of these and/or different shapes, etc.).

(27) A schematic diagram of illustrative components that may be used in device **10** is shown in FIG. **2**. As shown in FIG. **2**, device **10** may include control circuitry **38**. Control circuitry **38** may include storage such as storage circuitry **30**. Storage circuitry **30** may include hard disk drive storage, nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc.

(28) Control circuitry **38** may include processing circuitry such as processing circuitry **32**. Processing circuitry **32** may be used to control the operation of device **10**. Processing circuitry **32** may include one or more processors such as microprocessors, microcontrollers, digital signal processors, host processors, baseband processor integrated circuits, application specific integrated circuits, graphics processing units, central processing units (CPUs), etc. Control circuitry **38** may

be configured to perform operations in device **10** using hardware (e.g., dedicated hardware or circuitry), firmware, and/or software. Software code for performing operations in device **10** may be stored on storage circuitry **30** (e.g., storage circuitry **30** may include non-transitory (tangible) computer readable storage media that stores the software code). The software code may sometimes be referred to as program instructions, software, data, instructions, or code. Software code stored on storage circuitry **30** may be executed by processing circuitry **32**.

(29) Control circuitry **38** may be used to run software on device **10** such as internet browsing applications, voice-over-internet-protocol (VOIP) telephone call applications, email applications, media playback applications, operating system functions, etc. To support interactions with external equipment, control circuitry **38** may be used in implementing communications protocols. Communications protocols that may be implemented using control circuitry **38** include internet protocols, wireless local area network protocols (e.g., IEEE 802.11 protocols—sometimes referred to as WiFi®), protocols for other short-range wireless communications links such as the Bluetooth® protocol or other WPAN protocols, IEEE 802.11ad protocols, cellular telephone protocols, MIMO protocols, antenna diversity protocols, satellite navigation system protocols, antenna-based spatial ranging protocols (e.g., radio detection and ranging (RADAR) protocols or other desired range detection protocols for signals conveyed at millimeter and centimeter wave frequencies), etc. Each communication protocol may be associated with a corresponding radio access technology (RAT) that specifies the physical connection methodology used in implementing the protocol.

(30) Device **10** may include input-output circuitry **26**. Input-output circuitry **26** may include input-output devices **28**. Input-output devices **28** may be used to allow data to be supplied to device **10** and to allow data to be provided from device **10** to external devices. Input-output devices **28** may include user interface devices, data port devices, sensors, and other input-output components. For example, input-output devices **28** may include touch screens, displays without touch sensor capabilities, buttons, joysticks, scrolling wheels, touch pads, key pads, keyboards, microphones, cameras, speakers, status indicators, light sources, audio jacks and other audio port components, digital data port devices, light sensors, gyroscopes, accelerometers or other components that can detect motion and device orientation relative to the Earth, capacitance sensors, proximity sensors (e.g., a capacitive proximity sensor and/or an infrared proximity sensor), magnetic sensors, and other sensors and input-output components. The sensors in input-output devices **28** may include front-facing sensors that gather sensor data through display **14**. The front-facing sensors may be optical sensors. The optical sensors may include an image sensor (e.g., a front-facing camera), an infrared sensor, and/or an ambient light sensor. The infrared sensor may include one or more infrared emitters (e.g., a dot projector and a flood illuminator) and/or one or more infrared image sensors.

(31) Input-output circuitry **26** may include wireless circuitry such as wireless circuitry **34** for wirelessly conveying radio-frequency signals. While control circuitry **38** is shown separately from wireless circuitry **34** in the example of FIG. 2 for the sake of clarity, wireless circuitry **34** may include processing circuitry that forms a part of processing circuitry **32** and/or storage circuitry that forms a part of storage circuitry **30** of control circuitry **38** (e.g., portions of control circuitry **38** may be implemented on wireless circuitry **34**). As an example, control circuitry **38** may include baseband processor circuitry or other control components that form a part of wireless circuitry **34**.

(32) Wireless circuitry **34** may include radio-frequency (RF) transceiver circuitry formed from one or more integrated circuits, power amplifier circuitry, low-noise input amplifiers, passive RF components, one or more antennas, transmission lines, and other circuitry for handling RF wireless signals. Wireless signals can also be sent using light (e.g., using infrared communications).

(33) Wireless circuitry **34** may include radio-frequency transceiver circuitry **36** for handling transmission and/or reception of radio-frequency signals within corresponding frequency bands at radio frequencies (sometimes referred to herein as communications bands or simply as “bands”).

The frequency bands handled by radio-frequency transceiver circuitry **36** may include wireless local area network (WLAN) frequency bands (e.g., Wi-Fi® (IEEE 802.11) or other WLAN communications bands) such as a 2.4 GHz WLAN band (e.g., from 2400 to 2480 MHz), a 5 GHz WLAN band (e.g., from 5180 to 5825 MHz), a Wi-Fi® 6E band (e.g., from 5925-7125 MHz), and/or other Wi-Fi® bands (e.g., from 1875-5160 MHz), wireless personal area network (WPAN) frequency bands such as the 2.4 GHz Bluetooth® band or other WPAN communications bands, cellular telephone communications bands such as a cellular low band (LB) (e.g., 600 to 960 MHz), a cellular low-midband (LMB) (e.g., 1400 to 1550 MHz), a cellular midband (MB) (e.g., from 1700 to 2200 MHz), a cellular high band (HB) (e.g., from 2300 to 2700 MHz), a cellular ultra-high band (UHB) (e.g., from 3300 to 5000 MHz, or other cellular communications bands between about 600 MHz and about 5000 MHz), 3G bands, 4G LTE bands, 3GPP 5G New Radio Frequency Range 1 (FR1) bands below 10 GHz, 3GPP 5G New Radio (NR) Frequency Range 2 (FR2) bands between 20 and 60 GHz, other centimeter or millimeter wave frequency bands between 10-300 GHz, near-field communications frequency bands (e.g., at 13.56 MHz), satellite navigation frequency bands such as the Global Positioning System (GPS) L1 band (e.g., at 1575 MHz), L2 band (e.g., at 1228 MHz), L3 band (e.g., at 1381 MHz), L4 band (e.g., at 1380 MHz), and/or L5 band (e.g., at 1176 MHz), a Global Navigation Satellite System (GLONASS) band, a BeiDou Navigation Satellite System (BDS) band, ultra-wideband (UWB) frequency bands that operate under the IEEE 802.15.4 protocol and/or other ultra-wideband communications protocols (e.g., a first UWB communications band at 6.5 GHz and/or a second UWB communications band at 8.0 GHz), communications bands under the family of 3GPP wireless communications standards, communications bands under the IEEE 802.XX family of standards, satellite communications bands such as an L-band, S-band (e.g., from 2-4 GHz), C-band (e.g., from 4-8 GHz), X-band, Ku-band (e.g., from 12-18 GHz), Ka-band (e.g., from 26-40 GHz), etc., industrial, scientific, and medical (ISM) bands such as an ISM band between around 900 MHz and 950 MHz or other ISM bands below or above 1 GHz, one or more unlicensed bands, one or more bands reserved for emergency and/or public services, and/or any other desired frequency bands of interest. Wireless circuitry **34** may also be used to perform spatial ranging operations if desired.

(34) The UWB communications handled by radio-frequency transceiver circuitry **36** may be based on an impulse radio signaling scheme that uses band-limited data pulses. Radio-frequency signals in the UWB frequency band may have any desired bandwidths such as bandwidths between 499 MHz and 1331 MHz, bandwidths greater than 500 MHz, etc. The presence of lower frequencies in the baseband may sometimes allow ultra-wideband signals to penetrate through objects such as walls. In an IEEE 802.15.4 system, for example, a pair of electronic devices may exchange wireless time stamped messages. Time stamps in the messages may be analyzed to determine the time of flight of the messages and thereby determine the distance (range) between the devices and/or an angle between the devices (e.g., an angle of arrival of incoming radio-frequency signals).

(35) Radio-frequency transceiver circuitry **36** may include respective transceivers (e.g., transceiver integrated circuits or chips) that handle each of these frequency bands or any desired number of transceivers that handle two or more of these frequency bands. In scenarios where different transceivers are coupled to the same antenna, filter circuitry (e.g., duplexer circuitry, diplexer circuitry, low pass filter circuitry, high pass filter circuitry, band pass filter circuitry, band stop filter circuitry, etc.), switching circuitry, multiplexing circuitry, or any other desired circuitry may be used to isolate radio-frequency signals conveyed by each transceiver over the same antenna (e.g., filtering circuitry or multiplexing circuitry may be interposed on a radio-frequency transmission line shared by the transceivers). Radio-frequency transceiver circuitry **36** may include one or more integrated circuits (chips), integrated circuit packages (e.g., multiple integrated circuits mounted on a common printed circuit in a system-in-package device, one or more integrated circuits mounted on different substrates, etc.), power amplifier circuitry, up-conversion circuitry, down-conversion circuitry, low-noise input amplifiers, passive radio-frequency components, switching circuitry,

transmission line structures, and other circuitry for handling radio-frequency signals and/or for converting signals between radio-frequencies, intermediate frequencies, and/or baseband frequencies.

(36) In general, radio-frequency transceiver circuitry **36** may cover (handle) any desired frequency bands of interest. As shown in FIG. 2, wireless circuitry **34** may include antennas **40**. Radio-frequency transceiver circuitry **36** may convey radio-frequency signals using one or more antennas **40** (e.g., antennas **40** may convey the radio-frequency signals for the transceiver circuitry). The term “convey radio-frequency signals” as used herein means the transmission and/or reception of the radio-frequency signals (e.g., for performing unidirectional and/or bidirectional wireless communications with external wireless communications equipment). Antennas **40** may transmit the radio-frequency signals by radiating the radio-frequency signals into free space (or to freespace through intervening device structures such as a dielectric cover layer). Antennas **40** may additionally or alternatively receive the radio-frequency signals from free space (e.g., through intervening devices structures such as a dielectric cover layer). The transmission and reception of radio-frequency signals by antennas **40** each involve the excitation or resonance of antenna currents on an antenna resonating element in the antenna by the radio-frequency signals within the frequency band(s) of operation of the antenna.

(37) Antennas **40** in wireless circuitry **34** may be formed using any suitable antenna structures. For example, antennas **40** may include antennas with resonating elements that are formed from stacked patch antenna structures, loop antenna structures, patch antenna structures, inverted-F antenna structures, slot antenna structures, planar inverted-F antenna structures, waveguide structures, monopole antenna structures, dipole antenna structures, helical antenna structures, Yagi (Yagi-Uda) antenna structures, hybrids of these designs, etc. If desired, antennas **40** may include antennas with dielectric resonating elements such as dielectric resonator antennas. If desired, one or more of antennas **40** may be cavity-backed antennas. Two or more antennas **40** may be arranged in a phased antenna array if desired (e.g., for conveying centimeter and/or millimeter wave signals within a signal beam formed in a desired beam pointing direction that may be steered/adjusted over time). Different types of antennas may be used for different bands and combinations of bands.

(38) FIG. 3 is a schematic diagram showing how a given antenna **40** may be fed by radio-frequency transceiver circuitry **36**. As shown in FIG. 3, antenna **40** may have a corresponding antenna feed **50**. Antenna **40** may include one or more antenna resonating (radiating) elements **45** and an antenna ground **49**. Antenna resonating element(s) **45** may include one or more radiating arms, slots, waveguides, dielectric resonators, patches, parasitic elements, indirect feed elements, and/or any other desired antenna radiators. Antenna feed **50** may include a positive antenna feed terminal **52** coupled to at least one antenna resonating element **45** and a ground antenna feed terminal **44** coupled to antenna ground **49**. If desired, one or more conductive paths (sometimes referred to herein as ground paths, short paths, or return paths) may couple antenna resonating element(s) **45** to antenna ground **49**.

(39) Radio-frequency transceiver (TX/RX) circuitry **36** may be coupled to antenna feed **50** using a radio-frequency transmission line path **42** (sometimes referred to herein as transmission line path **42**). Transmission line path **42** may include a signal conductor such as signal conductor **46** (e.g., a positive signal conductor). Transmission line path **42** may include a ground conductor such as ground conductor **48**. Ground conductor **48** may be coupled to ground antenna feed terminal **44** of antenna feed **50**. Signal conductor **46** may be coupled to positive antenna feed terminal **52** of antenna feed **50**.

(40) Transmission line path **42** may include one or more radio-frequency transmission lines. The radio-frequency transmission line(s) in transmission line path **42** may include stripline transmission lines (sometimes referred to herein simply as striplines), coaxial cables, coaxial probes realized by metalized vias, microstrip transmission lines, edge-coupled microstrip transmission lines, edge-coupled stripline transmission lines, waveguide structures, combinations of these, etc. Multiple

types of radio-frequency transmission line may be used to form transmission line path **42**. Filter circuitry, switching circuitry, impedance matching circuitry, phase shifter circuitry, amplifier circuitry, and/or other circuitry may be interposed on transmission line path **42**, if desired. One or more antenna tuning components for adjusting the frequency response of antenna **40** in one or more bands may be interposed on transmission line path **42** and/or may be integrated within antenna **40** (e.g., coupled between the antenna ground and the antenna resonating element of antenna **40**, coupled between different portions of the antenna resonating element of antenna **40**, etc.).

(41) If desired, one or more of the radio-frequency transmission lines in transmission line path **42** may be integrated into ceramic substrates, rigid printed circuit boards, and/or flexible printed circuits. In one suitable arrangement, the radio-frequency transmission lines may be integrated within multilayer laminated structures (e.g., layers of a conductive material such as copper and a dielectric material such as a resin that are laminated together without intervening adhesive) that may be folded or bent in multiple dimensions (e.g., two or three dimensions) and that maintain a bent or folded shape after bending (e.g., the multilayer laminated structures may be folded into a particular three-dimensional shape to route around other device components and may be rigid enough to hold its shape after folding without being held in place by stiffeners or other structures). All the multiple layers of the laminated structures may be batch laminated together (e.g., in a single pressing process) without adhesive (e.g., as opposed to performing multiple pressing processes to laminate multiple layers together with adhesive).

(42) If desired, conductive electronic device structures such as conductive portions of housing **12** (FIG. **1**) may be used to form at least part of one or more of the antennas **40** in device **10**. FIG. **4** is a cross-sectional side view of device **10**, showing illustrative conductive electronic device structures that may be used in forming one or more of the antennas **40** in device **10**.

(43) As shown in FIG. **4**, peripheral conductive housing structures **12W** may extend around the lateral periphery of device **10** (e.g., as measured in the X-Y plane of FIG. **1**). Peripheral conductive housing structures **12W** may extend from rear housing wall **12R** (e.g., at the rear face of device **10**) to display **14** (e.g., at the front face of device **10**). In other words, peripheral conductive housing structures **12W** may form conductive sidewalls for device **10**, a first of which is shown in the cross-sectional side view of FIG. **4** (e.g., a given sidewall that runs along an edge of device **10** and that extends across the width or length of device **10**).

(44) Display **14** may have a display module such as display module **62** (sometimes referred to as a display panel). Display module **62** may include pixel circuitry, touch sensor circuitry, force sensor circuitry, and/or any other desired circuitry for forming active area AA of display **14**. Display **14** may include a dielectric cover layer such as display cover layer **64** that overlaps display module **62**. Display cover layer **64** may include plastic, glass, sapphire, ceramic, and/or any other desired dielectric materials. Display module **62** may emit image light and may receive sensor input (e.g., touch and/or force sensor input) through display cover layer **64**. Display cover layer **64** and display **14** may be mounted to peripheral conductive housing structures **12W**. The lateral area of display **14** that does not overlap display module **62** may form inactive area IA of display **14**.

(45) As shown in FIG. **4**, rear housing wall **12R** may be mounted to peripheral conductive housing structures **12W** (e.g., opposite display **14**). Rear housing wall **12R** may include a conductive layer such as conductive support plate **58**. Conductive support plate **58** may extend across an entirety of the width of device **10** (e.g., between the left and right edges of device **10** as shown in FIG. **1**). Conductive support plate **58** may be formed from an integral portion of peripheral conductive housing structures **12W** that extends across the width of device **10** or may include a separate housing structure attached, coupled, or affixed to peripheral conductive housing structures **12W**.

(46) If desired, rear housing wall **12R** may include a dielectric cover layer such as dielectric cover layer **56**. Dielectric cover layer **56** may include glass, plastic, sapphire, ceramic, one or more dielectric coatings, or other dielectric materials. Dielectric cover layer **56** may be layered under conductive support plate **58** (e.g., conductive support plate **58** may be coupled to an interior surface

of dielectric cover layer 56). If desired, dielectric cover layer 56 may extend across an entirety of the width of device 10 and/or an entirety of the length of device 10. Dielectric cover layer 56 may overlap slot 60. If desired, dielectric cover layer 56 be provided with pigmentation and/or an opaque masking layer (e.g., an ink layer) that helps to hide the interior of device 10 from view. In another suitable arrangement, dielectric cover layer 56 may be omitted and slot 60 may be filled with a solid dielectric material.

(47) The housing for device 10 may also include one or more additional conductive support plates interposed between display 14 and rear housing wall 12R. For example, the housing for device 10 may include a conductive support plate such as mid-chassis 65 (sometimes referred to herein as conductive support plate 65). Mid-chassis 65 may be vertically interposed between rear housing wall 12R and display 14 (e.g., conductive support plate 58 may be located at a first distance from display 14 whereas mid-chassis 65 is located at a second distance that is less than the first distance from display 14). Mid-chassis 65 may extend across an entirety of the width of device 10 (e.g., between the left and right edges of device 10 as shown in FIG. 1). Mid-chassis 65 may be formed from an integral portion of peripheral conductive housing structures 12W that extends across the width of device 10 or may include a separate housing structure attached, coupled, or affixed to peripheral conductive housing structures 12W. One or more components may be supported by mid-chassis 65 (e.g., logic boards such as a main logic board, a battery, etc.) and/or mid-chassis 65 may contribute to the mechanical strength of device 10. Mid-chassis 65 may be formed from metal (e.g., stainless steel, aluminum, etc.).

(48) Conductive support plate 58, mid-chassis 65, and/or display module 62 may have an edge 54 that is separated from peripheral conductive housing structures 12W by dielectric-filled slot 60 (sometimes referred to herein as opening 60, gap 60, or aperture 60). Slot 60 may be filled with air, plastic, ceramic, or other dielectric materials. Conductive housing structures such as conductive support plate 58, mid-chassis 65, conductive portions of display module 62, and/or peripheral conductive housing structures 12W (e.g., the portion of peripheral conductive housing structures 12W opposite conductive support plate 58, mid-chassis 65, and display module 62 at slot 60) may be used to form antenna structures for one or more of the antennas 40 in device 10.

(49) For example, peripheral conductive housing structures 12W may form an antenna resonating element arm (e.g., an inverted-F antenna resonating element arm) in the antenna resonating element 45 of an antenna 40 in device 10. Mid-chassis 65, conductive support plate 58, and/or display module 62 may be used to form the antenna ground 49 (FIG. 3) for one or more of the antennas 40 in device 10 and/or to form one or more edges of slot antenna resonating elements for the antennas in device 10. One or more conductive interconnect structures 63 may electrically couple mid-chassis 65 to conductive support plate 58 and/or one or more conductive interconnect structures 63 may electrically couple mid-chassis 65 to conductive structures in display module 62 (sometimes referred to herein as conductive display structures) so that each of these elements form part of the antenna ground. The conductive display structures may include a conductive frame, bracket, or support for display module 62, shielding layers in display module 62, ground traces in display module 62, etc.

(50) Conductive interconnect structures 63 may serve to ground mid-chassis 65 to conductive support plate 58 and/or display module 62 (e.g., to ground conductive support plate 58 to the conductive display structures through mid-chassis 65). Put differently, conductive interconnect structures 63 may hold the conductive display structures, mid-chassis 65, and/or conductive support plate 58 to a common ground or reference potential (e.g., as a system ground for device 10 that is used to form part of antenna ground 49 of FIG. 3). Conductive interconnect structures 63 may therefore sometimes be referred to herein as grounding structures 63, grounding interconnect structures 63, or vertical grounding structures 63. Conductive interconnect structures 63 may include conductive traces, conductive pins, conductive springs, conductive prongs, conductive brackets, conductive screws, conductive clips, conductive tape, conductive wires, conductive

traces, conductive foam, conductive adhesive, solder, welds, metal members (e.g., sheet metal members), contact pads, conductive vias, conductive portions of one or more components mounted to mid-chassis **65** and/or conductive support plate **58**, and/or any other desired conductive interconnect structures.

(51) If desired, device **10** may include multiple slots **60** and peripheral conductive housing structures **12W** may include multiple dielectric gaps that divide the peripheral conductive housing structures into segments (e.g., dielectric gaps **18** of FIG. **1**). FIG. **5** is a top interior view showing how the lower end of device **10** (e.g., within region **22** of FIG. **1**) may include a slot **60** and may include multiple dielectric gaps that divide the peripheral conductive housing structures into segments for forming multiple antennas. Display **14** and other internal components have been removed from the view shown in FIG. **5** for the sake of clarity.

(52) As shown in FIG. **5**, peripheral conductive housing structures **12W** may include a first conductive sidewall at the left edge of device **10**, a second conductive sidewall at the top edge of device **10** (not shown in FIG. **5**), a third conductive sidewall at the right edge of device **10**, and a fourth conductive sidewall at the bottom edge of device **10** (e.g., in an example where device **10** has a substantially rectangular lateral shape). Peripheral conductive housing structures **12W** may be segmented by dielectric-filled gaps **18** such as a first gap **18-1**, a second gap **18-2**, and a third gap **18-3**. Gaps **18-1**, **18-2**, and **18-3** may be filled with plastic, ceramic, sapphire, glass, epoxy, or other dielectric materials. The dielectric material in the gaps may lie flush with peripheral conductive housing structures **12W** at the exterior surface of device **10** if desired.

(53) Gap **18-1** may divide the first conductive sidewall to separate segment **66** of peripheral conductive housing structures **12W** from segment **68** of peripheral conductive housing structures **12W**. Gap **18-2** may divide the third conductive sidewall to separate segment **72** from segment **70** of peripheral conductive housing structures **12W**. Gap **18-3** may divide the fourth conductive sidewall to separate segment **68** from segment **70** of peripheral conductive housing structures **12W**. In this example, segment **68** forms the bottom-left corner of device **10** (e.g., segment **68** may have a bend at the corner) and is formed from the first and fourth conductive sidewalls of peripheral conductive housing structures **12W** (e.g., in lower region **22** of FIG. **1**). Segment **70** forms the bottom-right corner of device **10** (e.g., segment **70** may have a bend at the corner) and is formed from the third and fourth conductive sidewalls of peripheral conductive housing structures **12W** (e.g., in lower region **22** of FIG. **1**).

(54) Device **10** may include ground structures **78** (e.g., structures that form part of the antenna ground for one or more of the antennas in device **10**). Ground structures **78** may include one or more metal layers such as conductive support plate **58** (FIG. **4**), mid-chassis **65** (FIG. **4**), conductive display structures, conductive interconnect structures **63** (FIG. **4**), conductive traces on a printed circuit board, conductive portions of one or more components in device **10**, etc. Ground structures **78** may extend between opposing sidewalls of peripheral conductive housing structures **12W**. For example, ground structures **78** may extend from segment **66** to segment **72** of peripheral conductive housing structures **12W** (e.g., across the width of device **10**, parallel to the X-axis of FIG. **5**). Ground structures **78** may be welded or otherwise affixed to segments **66** and **72**. In another suitable arrangement, some or all of ground structures **78**, segment **66**, and segment **72** may be formed from a single, integral (continuous) piece of machined metal (e.g., in a unibody configuration). Device **10** may have a longitudinal axis **76** that bisects the width of device **10** and that runs parallel to the length of device **10** (e.g., parallel to the Y-axis).

(55) As shown in FIG. **5**, slot **60** may separate ground structures **78** from segments **68** and **70** of peripheral conductive housing structures **12W** (e.g., the upper edge of slot **60** may be defined by ground structures **78** whereas the lower edge of slot **60** is defined by segments **68** and **70**). Slot **60** may have an elongated shape extending from a first end at gap **18-1** to an opposing second end at gap **18-2** (e.g., slot **60** may span the width of device **10**). Slot **60** may be filled with air, plastic, glass, sapphire, epoxy, ceramic, or other dielectric material. Slot **60** may be continuous with gaps

18-1, 18-2, and 18-3 in peripheral conductive housing structures **12W** if desired (e.g., a single piece of dielectric material may be used to fill both slot **60** and gaps **18-1, 18-2, and 18-3**).

(56) Ground structures **78**, segment **66**, segment **68**, segment **70**, and portions of slot **60** may be used in forming multiple antennas **40** in the lower region of device **10** (sometimes referred to herein as lower antennas). For example, device **10** may include a first antenna **40-1** having an antenna resonating (radiating) element formed from segment **66**, segment **68**, and/or a portion of slot **60** (e.g., a vertically extending end of slot **60** that extends parallel to longitudinal axis **76** and past gap **18-1**, between segment **66** and ground structures **78**) and having an antenna ground formed from ground structures **78**. Device **10** may also include a second antenna **40-2** having an antenna resonating element formed from segment **68**, segment **66**, and/or a portion of slot **60** and having an antenna ground formed from ground structures **78**. Device **10** may also include a third antenna **40-3** having an antenna resonating element (e.g., a resonating element arm) formed from segment **70** and having an antenna ground formed from ground structures **78**. Device **10** may also include a fourth antenna **40-4** having a slot antenna resonating element formed from segment **72** and/or a portion of slot **60** between segment **72** and ground structures **78**. Antennas **40-2** and **40-3** may be, for example, inverted-F antennas having return paths that couples the respective resonating element arms to the antenna ground. If desired, device **10** may also include a fifth antenna **40-5** at least partially overlapping the volume of antenna **40-2**. Fifth antenna **40-5** may include an antenna resonating element arm formed from conductive traces on a flexible printed circuit and may have an antenna ground formed from ground structures **78**. Antennas **40-1, 40-2, 40-3, 40-4, and 40-5** may convey radio-frequency signals in one or more frequency bands (e.g., using MIMO communications in one or more of bands, thereby maximizing data throughput).

(57) It may be desirable for each antenna **40** to cover as many frequency bands as possible (thereby maximizing the wireless capability of device **10**) in as small a volume as possible (thereby minimizing the size of device **10**). In practice, conductive structures in a given antenna **40** (e.g., antenna resonating element **45** and/or antenna ground **49** of FIG. 3) may be used to boost the wireless performance of an adjacent antenna **40** via near-field (NF) electromagnetic coupling. FIG. 6 is a diagram showing how an antenna **40** may be used to boost the wireless performance of one or more adjacent antennas **40**.

(58) As shown in FIG. 6, device **10** may include multiple antennas **40**. Each antenna **40** may be fed using a respective antenna feed **50**. Each antenna **40** may be fed by a transmission line path **42** coupled to its antenna feed **50**. Each antenna **40** may have a frequency response that is tuned or adjusted by a respective tuner **80** disposed on its transmission line path **42**. Tuner **80** may, for example, perform impedance matching and/or aperture tuning for its antenna **40**.

(59) At the same time, a given antenna **40** may have conductive structures (e.g., antenna resonating element **45** and/or antenna ground **49** of FIG. 3) that are wirelessly coupled to conductive structures in one, two, or more than two nearby (e.g., adjacent) antennas **40** via near-field electromagnetic coupling(s) **82**. Near-field electromagnetic coupling(s) **82** may allow a given antenna **40** to boost the response (e.g., increase antenna efficiency) in one or more frequency bands for the adjacent antenna(s) **40**. This may, for example serve, to increase the bandwidth of the adjacent antenna(s) **40** (e.g., by increasing the range of frequencies over which the adjacent antenna(s) **40** exhibit an antenna efficiency that exceeds a threshold value). Conversely, near-field electromagnetic coupling(s) **82** may allow a given antenna **40** to have its own response boosted across one or more frequency bands by the presence of the conductive structures in the adjacent antenna(s) **40**.

(60) If desired, two or more antennas **40** may share conductive structures. For example, the antenna resonating element **45** (FIG. 3) of a first antenna **40** may include one or more of the same (shared) conductive structures of the antenna resonating element **45** of a second antenna **40** (e.g., the antenna feed **50** for both the first and second antennas may be coupled to the same conductive structure(s)). Sharing conductive structures between the antenna resonating element **45** of a given

antenna **40** and one or more adjacent antenna(s) **40** and/or near-field electromagnetic coupling(s) **82** may allow the tuner **80** of a given antenna **40** to also tune the frequency response of one or more adjacent antenna(s) **40**. In other words, tuner **80** may effectively boost the antenna efficiency and/or bandwidth of its own corresponding antenna **40** in a first frequency band (or set of frequency bands) covered by that antenna **40** while concurrently boosting the antenna efficiency and/or bandwidth of one or more adjacent antenna(s) **40** in a second frequency band (or set of frequency bands) covered by the one or more adjacent antenna(s) **40**.

(61) For example, the tuner **80** interposed on the transmission line path **42** of a first antenna **40** may perform impedance matching for first antenna **40**. In other words, the tuner may form impedance matching circuitry for the first antenna that serves to match an impedance of transmission line path **42** to the impedance of first antenna **40** in a first set of frequency bands handled by the first antenna. Tuner **80** may also concurrently perform aperture tuning for a second antenna **40** adjacent to the first antenna. In other words, the tuner may form an aperture tuner for the second antenna, which serves to tune a frequency response of the second antenna in a second set of frequency bands handled by the second antenna. The second antenna **40** and the first antenna **40** may both include antenna resonating elements formed from one or more of the same (shared) conductive structures, such as the same segment of peripheral conductive housing structures **12W** and/or the same portion of slot **60** (FIG. 5).

(62) Tuners **80** may each include one or more tuning components (e.g., tuning elements or tuning circuits). The tuning components may include one or more fixed or adjustable capacitors, inductors, and/or resistors, switching circuitry (e.g., one or more switches), one or more filters, one or more radio-frequency couplers, and/or any other desired tuning components coupled in any desired manner between terminals of the tuner. The tuning components may, if desired, be covered with an encapsulation layer or cap and/or an electromagnetic shield. The tuning components may be mounted to an underlying substrate such as a flexible printed circuit.

(63) FIG. 7 is an interior rear view showing how antennas **40-2** and **40-1** may both include antenna resonating elements formed from one or more of the same (shared) conductive structures. Rear housing wall **12R** (FIG. 4) and other components in device **10** have been omitted from the view shown in FIG. 7 for the sake of clarity.

(64) As shown in FIG. 7, antenna **40-1** and antenna **40-2** may both have an antenna resonating element arm (e.g., antenna resonating element **45** of FIG. 3) formed from segment **68** of peripheral conductive housing structures **12W**. Antennas **40-1** and **40-2** may also share ground structures **78**. Antennas **40-1** and **40-2** may include a return path **86** coupled between terminal **88** on segment **68** (e.g., at or adjacent to gap **18-3**) and terminal **90** on ground structures **78**. If desired, a tuning component for antenna **40-1** and/or **40-2** may be disposed on return path **86**.

(65) Antenna **40-1** may be fed using a first antenna feed **50-1** coupled across slot **60**. Antenna feed **50-1** may have a positive antenna feed terminal **52** coupled to a first location on segment **68** and may have a ground antenna feed terminal **44** coupled to ground structures **78**. Antenna **40-2** may be fed using a first antenna feed **50-2** coupled across slot **60**. Antenna feed **50-2** may have a positive antenna feed terminal **52** coupled to a second location on segment **68** and may have a ground antenna feed terminal **44** coupled to ground structures **78**. The first location on segment **68** (e.g., positive antenna feed terminal **52** of antenna feed **50-1**) may be interposed between gap **18-1** and the second location on segment **68** (e.g., positive antenna feed terminal **52** of antenna feed **50-2**). In other words, antenna feed **50-1** may be interposed between gap **18-1** and antenna feed **50-2**.

(66) Antenna **40-1** may be fed by a first transmission line path **42-1** coupled to antenna feed **50-1**. First transmission line path **42-1** may have a signal conductor **46** coupled to the positive antenna feed terminal **52** of antenna feed **50-1** and may have a ground conductor **48** coupled to the ground antenna feed terminal **44** of antenna feed **50-1**. Antenna **40-2** may be fed by a second transmission line path **42-2** coupled to antenna feed **50-2**. Second transmission line path **42-2** may have a signal conductor **46** coupled to the positive antenna feed terminal **52** of antenna feed **50-2** and may have a

ground conductor **48** coupled to the ground antenna feed terminal **44** of antenna feed **50-2**.

Transmission line paths **42-1** and **42-2** may be coupled to one or more transceivers **36** (FIG. 3).

(67) Antenna **40-1** may include a tuner **80-1** disposed on transmission line path **42-1** (e.g., having a first terminal coupled to the positive antenna feed terminal **52** and having a second terminal coupled to the ground antenna feed terminal **44** of antenna feed **50-1**). Antenna **40-2** may include a tuner **80-2** disposed on transmission line path **42-2** (e.g., coupled between the positive antenna feed terminal **52** and the ground antenna feed terminal **44** of antenna feed **50-2**). Tuners **80-1** and **80-2** may receive control signals that adjust the state of one or more tuning components in tuners **80-1** and **80-2** (e.g., to alter the frequency response of antenna **40-1** and/or antenna **40-2** as needed over time).

(68) Slot **60** may include a vertical portion that extends parallel to longitudinal axis **76** of FIG. 5 (Y-axis of FIG. 7) and beyond gap **18-1**. As shown in FIG. 7, slot **60** may include an extended (elongated) portion **84**. Extended portion **84** of slot **60** may extend between segment **66** and ground structures **78** (e.g., segment **66** and ground structures **78** may define opposing edges of extended portion **84**), parallel to the Y-axis. Extended portion **84** of slot **60** may have an open end at gap **18-1** and an opposing closed end formed from ground structures **78**. Extended portion **84** of slot **60** may sometimes be referred to herein simply as slot **84**.

(69) The length of segment **68** may be selected to contribute to the radiative response (resonance) of antenna **40-1** in a first set of frequency bands while also contributing to the radiative response of antenna **40-2** in a second set of frequency bands (e.g., in a fundamental mode and/or one or more harmonic modes). The first set of frequency bands covered by antenna **40-1** may include the cellular UHB (e.g., frequencies between about 3400 MHz and 3600 MHz) whereas the second set of frequency bands covered by antenna **40-2** may include the cellular MB, the cellular HB, and the 2.4 GHz WLAN/WPAN band, for example.

(70) Slot **84** may also contribute to the radiative response of antenna **40-1** in the first set of frequency bands as well as the radiative response of antenna **40-2** in the second set of frequency bands. If desired, slot **84** may form a slot antenna resonating element for both antennas **40-1** and **40-2**. (FIG. 5). Slot **84** may be near-field coupled to the segment **68** (e.g., the radiating arm of both antennas **40-1** and **40-2**) via near-field electromagnetic coupling **82**. For example, antenna currents conveyed by antenna feed **50-1** in the first set of frequency bands and flowing on segment **68** may produce (induce), via near-field electromagnetic coupling **82**, corresponding antenna currents that flow around the perimeter of slot **84** (e.g., ground structures **78** and segment **66**). These antenna currents and thus slot **84** may be used to convey radio-frequency signals for antenna feed **50-1** and antenna **40-1** in the first set of frequency bands (e.g., slot **84** may contribute to the resonance of antenna **40-1**). At the same time, antenna currents conveyed by antenna feed **50-2** in the second set of frequency bands and flowing on segment **68** may produce (induce), via near-field electromagnetic coupling **82**, corresponding antenna currents that flow around the perimeter of slot **84** (e.g., ground structures **78** and segment **66**). These antenna currents and thus slot **84** may be used to convey radio-frequency signals for antenna feed **50-1** and antenna **40-1** in the second set of frequency bands (e.g., slot **84** may contribute to the resonance of antenna **40-2**).

(71) Antennas **40-1** and **40-2** may also include a tuner **102** coupled between terminal **100** on ground structures **78** and terminal **98** on segment **66** across slot **84** (e.g., tuner **102** may bridge slot **84**). Near-field coupling **82** may allow tuner **102** to adjust (tune) the radiative response of both antenna **40-1** (in the first set of frequency bands) and antenna **40-2** (in the second set of frequency bands). Tuner **102** may receive control signals that change the state of tuner **102** and thus the frequency response of antenna **40-1** and/or antenna **40-2**. Tuner **102** may include one or more tuning components (e.g., tuning elements or tuning circuits). The tuning components may include one or more fixed or adjustable capacitors, inductors, and/or resistors, switching circuitry (e.g., one or more switches), one or more filters, one or more radio-frequency couplers, and/or any other desired tuning components coupled in any desired manner between terminals of the tuner. The

tuning components may, if desired, be covered with an encapsulation layer or cap and/or an electromagnetic shield. The tuning components may be mounted to an underlying substrate such as a flexible printed circuit.

(72) Tuner **80-1** of antenna **40-1** may contribute to the radiative response of both antennas **40-1** and **40-2**. Tuner **80-1** may, for example, form impedance matching circuitry for antenna **40-1** that performs impedance matching in the first set of frequency bands between transmission line path **42-1** and segment **68**. At the same time, tuner **80-1** may form an aperture tuner that tunes the frequency response of antenna **40-2** in the second set of frequency bands. The tuning components in tuner **80-1** may, for example, be selected to perform impedance matching in the first set of frequency bands while concurrently adjusting the impedance between segment **68** and ground structures **78** (across slot **60**) in the second set of frequency bands for tuning antenna **40-2**.

(73) Similarly, tuner **80-2** of antenna **40-2** may contribute to the radiative response of both antennas **40-1** and **40-2**. Tuner **80-2** may, for example, form impedance matching circuitry that performs impedance matching for antenna **40-2** in the second set of frequency bands between transmission line path **42-2** and segment **68**. At the same time, tuner **80-2** may form an aperture tuner that tunes the frequency response of antenna **40-2** in the second set of frequency bands. The tuning components in tuner **80-2** may, for example, be selected to perform impedance matching in the second set of frequency bands while concurrently adjusting the impedance load between segment **68** and ground structures **78** (across slot **60**) in the first set of frequency bands for tuning antenna **40-1**. Tuners **80-1** and **80-2**, tuner **102**, slot **84**, and the length of segment **68** may collectively configure the conductive structures in FIG. 7 (e.g., the antenna resonating element of antennas **40-1** and **40-2**) to radiate and convey radio-frequency signals across a relatively wide frequency range that includes both the first and second sets of frequency bands (e.g., from 1.7-6 GHz), despite the small volume in device **10** occupied by antennas **40-1** and **40-2**.

(74) As shown in FIG. 7, antenna **40-5** may also be disposed in device **10** adjacent to antennas **40-1** and **40-2**. Antenna **40-5** may have an antenna resonating element formed from conductive traces **96** (e.g., a resonating element arm) patterned on an underlying substrate such as substrate **94** (e.g., a flexible printed circuit board). Antenna **40-5** may be fed by antenna feed **50-3** having a positive antenna feed terminal **52** on conductive traces **96** and a ground antenna feed terminal **44** on ground structures **78**. Antenna **40-5** may be fed by a transmission line path **42-3** having a signal conductor **46** coupled to positive antenna feed terminal **52** of antenna feed **50-3** and having a ground conductor **48** coupled to ground antenna feed terminal **44** of antenna feed **50-3**. Antenna **40-5** may include a tuner **80-3** disposed on transmission line path **42-3** that adjusts a frequency response of antenna **40-5**. Antenna **40-5** may convey radio-frequency signals in a 5 GHz WLAN band and/or cellular telephone bands around 5 GHz.

(75) In the example of FIG. 7, tuner **80-3** of antenna **40-5** only tunes the frequency response of antenna **40-5** without tuning the frequency response of antennas **40-1** and **40-2**. If desired, antenna **40-5** may have an antenna resonating element formed from segment **66** and slot **84** and tuner **80-3** may be used to tune the frequency response of antennas **40-1** and **40-2**. FIG. 8 is a diagram showing how antenna **40-5** may have an antenna resonating element formed from segment **66** and slot **84** and tuner **80-3** may be used to tune the frequency response of antennas **40-1** and **40-2**.

(76) As shown in FIG. 8, antenna feed **50-3** may be coupled across slot **84**. Positive antenna feed terminal **52** of antenna feed **50-3** may be coupled to segment **66**. Ground antenna feed terminal **44** may be coupled to ground structures **78**. Tuner **80-3** may be coupled between positive antenna feed terminal **52** and ground antenna feed terminal **44** of antenna feed **50-3**. Segment **66** and slot **84** may form the antenna resonating element for antenna **40-5**. Antenna feed **50-4** may convey antenna currents around the perimeter of slot **84** in the 5 GHz WLAN band and/or cellular telephone bands around 5 GHz.

(77) Tuner **80-3** may perform impedance matching for antenna **40-5** and may tune the frequency response of antenna **40-5**. At the same time, tuner **80-3** may tune the frequency response of antenna

40-1 and/or antenna **40-2** via near-field electromagnetic coupling **82** and/or by adjusting the impedance across slot **84** for antenna current in the first set of frequency bands (e.g., for antenna **40-1**) and/or for antenna current in the second set of frequency bands (e.g., for antenna **40-2**). The examples of FIGS. **7** and **8** are merely illustrative. Segment **68** and slot **84** may have other shapes (e.g., having any desired number of curved and/or straight segments or portions). Additional tuners may be coupled to segment **68** and/or segment **66**. Segment **66**, segment **68**, slot **84**, ground structures **78**, return path **86**, tuners **80-1**, **80-2**, and **80-3**, and/or antenna feeds **50-1**, **50-2**, and **50-3** may sometimes be referred to collectively herein as conductive structures, conductive antenna structures, or antenna structures.

(78) Device **10** may gather and/or use personally identifiable information. It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

(79) The foregoing is merely illustrative and various modifications can be made by those skilled in the art without departing from the scope and spirit of the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. An electronic device comprising: ground structures; peripheral conductive housing structures having a dielectric-filled gap that divides the peripheral conductive housing structures into a first segment and a second segment, wherein the first segment is separated from the ground structures by a slot and the second segment is separated from the ground structures by a portion of the slot extending beyond the dielectric-filled gap; a first antenna feed coupled between the ground structures and a first location on the first segment across the slot, wherein the first antenna feed is configured to convey first radio-frequency signals in a first frequency band, the first segment and the portion of the slot being configured to radiate in the first frequency band; and a second antenna feed coupled between the ground structures and a second location on the first segment across the slot, wherein the second antenna feed is configured to convey second radio-frequency signals in a second frequency band different than the first frequency band, the first segment and the portion of the slot being configured to radiate in the second frequency band.
2. The electronic device of claim 1, wherein the first segment is near-field coupled to the portion of the slot in the first frequency band.
3. The electronic device of claim 2, wherein the first segment is near-field coupled to the portion of the slot in the second frequency band.
4. The electronic device of claim 1, further comprising: a tuner coupled between the second segment and the ground structures across the portion of the slot.
5. The electronic device of claim 4, wherein the tuner is configured to adjust the first frequency band and the second frequency band.
6. The electronic device of claim 1, further comprising: a first transmission line path coupled to the first antenna feed; a first tuner disposed on the first transmission line path; a second transmission line path coupled to the second antenna feed; and a second tuner disposed on the second transmission line path.
7. The electronic device of claim 6, wherein the first tuner is configured to impedance match the first transmission line path to the first segment in the first frequency band and the second tuner is configured to impedance match the second transmission line path to the first segment in the second frequency band.
8. The electronic device of claim 7, wherein the first tuner is further configured to adjust the second

frequency band and the second tuner is further configured to adjust the first frequency band.

9. The electronic device of claim 8, further comprising a third tuner coupled between the second segment and the ground structures across the portion of the slot.

10. The electronic device of claim 1, further comprising a return path that couples a third location on the first segment to the ground structures, the second location being interposed on the first segment between the third location and the first location.

11. An electronic device comprising: ground structures; peripheral conductive housing structures having a segment extending from a first dielectric-filled gap to a second dielectric-filled gap; a first antenna feed terminal coupled to a first location on the segment; a second antenna feed terminal coupled to the ground structures; a third antenna feed terminal coupled to a second location on the segment; a fourth antenna feed terminal coupled to the ground structures; a first transmission line path coupled to the first and second antenna feed terminals; a second transmission line path coupled to the third and fourth antenna feed terminals; a first tuner disposed on the first transmission line path, wherein the first tuner is configured to match an impedance of the first transmission line path to an impedance of the segment in a first frequency band; and a second tuner disposed on the second transmission line path, wherein the second tuner is configured to match an impedance of the second transmission line path to the impedance of the segment in a second frequency band different from the first frequency band, the second tuner being further configured to tune a resonance of the segment in the first frequency band.

12. The electronic device of claim 11, wherein the first tuner is further configured to tune a resonance of the segment in the second frequency band.

13. The electronic device of claim 12, wherein the peripheral conductive housing structures comprise an additional segment separated from the segment by the first dielectric-filled gap, the additional segment is separated from the ground structures by a slot, and the slot is near-field coupled to the segment in the first frequency band and in the second frequency band.

14. The electronic device of claim 13, further comprising: a third tuner coupled between the ground structures and the additional segment across the slot, the third tuner being configured to tune the resonance of the segment in the first frequency band and the resonance of the segment in the second frequency band.

15. The electronic device of claim 13, further comprising: a fifth antenna feed terminal coupled to the additional segment; a sixth antenna feed terminal coupled to the ground structures; a third transmission line path coupled to the fifth and sixth antenna feed terminals; and a third tuner disposed on the third transmission line path, wherein the third tuner is configured to match an impedance of the third transmission line path to an impedance of the additional segment in a third frequency band different from the first and second frequency bands, the third tuner being further configured to tune the resonance of the segment in the first frequency band and the resonance of the segment in the second frequency band.

16. The electronic device of claim 11, wherein the first tuner has a first terminal coupled to the first antenna feed terminal and has a second terminal coupled to the second antenna feed terminal and wherein the second tuner has a third terminal coupled to the third antenna feed terminal and has a fourth terminal coupled to the fourth antenna feed terminal.

17. The electronic device of claim 11, further comprising: a return path that couples a third location on the segment to the ground structures, the second location being interposed on the segment between the first location and the third location.

18. An electronic device comprising: peripheral conductive housing structures having a dielectric-filled gap that divides the peripheral conductive housing structures into a first segment and a second segment; ground structures separated from the first segment by a first slot and separated from the second segment by a second slot extending from the first slot; a first antenna feed coupled between the first segment and the ground structures across the first slot; a second antenna feed coupled between the second segment and the ground structures across the second slot, the second

antenna feed being configured to excite a resonance of the second segment in a first frequency band; a third antenna feed coupled between the second segment and the ground structures across the second slot, the third antenna feed being configured to excite a resonance of the second segment in a second frequency band different than the first frequency band; and a tuner coupled to the first antenna feed, wherein the tuner is configured to adjust the resonance of the segment in the first frequency band.

19. The electronic device of claim 18, wherein the tuner is configured to impedance match the third antenna feed in a third frequency band that is different from the first frequency band and the second frequency band.

20. The electronic device of claim 18, wherein the tuner is configured to adjust the resonance of the segment in the second frequency band.
